

UTSJOKI NUORGAM, Finland

WMO: 028250

Lat: 70.0819N

Lon: 27.8964E

Elev: 22

StdP: 101.06

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.8	-28.8	-35.4	0.1	-31.7	-32.0	0.2	-28.8	11.1	-5.0	9.3	-4.9	0.8	200	0.735

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	23.3	16.3	20.6	15.2	18.4	14.0	17.3	21.8	15.7	19.5	14.4	17.7	2.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	10.8	19.1	13.9	10.0	17.7	12.7	9.2	16.2	48.6	22.5	44.2	19.4	40.5	18.1	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.0	6.6	5.6	-34.6	26.1	3.9	2.7	-37.4	28.0	-39.7	29.6	-41.9	31.2	-44.7	33.2		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.3	-12.4	-11.3	-7.9	-2.0	4.6	8.8	12.4	10.4	7.1	0.2	-5.7	-8.9		
	DBStd	10.07	7.71	8.20	6.18	4.91	3.55	3.57	3.57	3.27	3.16	4.47	5.95	6.79		
	HDD10.0	3950	693	598	556	359	173	64	13	34	97	306	470	587		
	HDD18.3	6814	952	831	814	609	425	286	186	246	337	564	720	846		
	CDD10.0	182	0	0	0	0	6	29	89	47	9	0	0	0		
	CDD18.3	5	0	0	0	0	0	1	4	1	0	0	0	0		
	CDH23.3	85	0	0	0	0	6	7	63	9	0	0	0	0		
	CDH26.7	17	0	0	0	0	2	0	13	2	0	0	0	0		
	WSAvg	2.4	2.3	2.5	2.6	2.5	2.8	2.8	2.4	2.1	2.1	2.0	2.1	2.3		
	PrecAvg	448	30	29	29	21	25	57	53	48	35	48	33	36		
Precipitation	PrecMax	611	57	54	46	37	51	104	139	80	80	107	55	60		
	PrecMin	330	8	5	10	3	9	8	15	22	7	8	10	17		
	PrecStd	66	15	13	9	8	12	25	36	18	18	26	15	14		
	PrecStd	66	15	13	9	8	12	25	36	18	18	26	15	14		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	3.6	3.9	4.4	9.5	22.2	24.1	28.6	24.2	18.3	11.2	5.1	4.7			
	0.4% MCWB	2.0	2.3	2.3	6.0	12.8	16.3	18.9	16.7	14.2	8.9	4.2	2.9			
	2% DB	1.8	2.4	3.1	6.5	16.9	20.4	25.0	21.0	16.3	9.3	3.7	2.4			
	2% MCWB	0.5	1.1	1.4	3.7	10.2	14.1	17.4	15.8	12.7	7.6	2.7	1.3			
	5% DB	-0.6	0.8	1.8	5.1	13.5	17.7	21.9	18.8	14.5	7.6	2.5	1.4			
	5% MCWB	-1.6	-0.4	0.3	2.7	8.6	12.8	16.1	14.4	11.5	6.5	1.6	0.4			
	10% DB	-2.3	-0.9	0.6	4.1	10.8	15.3	19.2	16.5	12.7	6.2	1.5	0.2			
	10% MCWB	-3.1	-1.8	-0.7	2.1	6.8	11.2	15.0	13.3	10.5	5.3	0.7	-0.7			
	0.4% WB	2.2	2.5	2.5	6.1	13.8	17.0	19.6	18.1	14.8	9.1	4.6	3.0			
	0.4% MCDB	3.4	3.6	4.2	9.4	21.1	22.0	26.0	22.0	17.5	10.9	5.1	4.6			
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	0.6	1.2	1.5	3.8	10.4	15.2	18.2	16.2	13.3	7.8	2.8	1.5			
	2% MCDB	1.7	2.3	2.9	5.9	15.8	19.0	23.6	20.0	15.9	9.2	3.6	2.3			
	5% WB	-1.5	-0.3	0.4	3.0	8.8	13.3	16.6	14.9	11.8	6.6	1.7	0.5			
	5% MCDB	-0.7	0.7	1.7	4.9	13.2	17.4	21.0	17.8	13.8	7.5	2.4	1.3			
	10% WB	-3.1	-1.8	-0.7	2.3	7.3	11.5	15.1	13.7	10.7	5.3	0.7	-0.7			
	10% MCDB	-2.4	-0.9	0.6	3.9	10.3	14.8	18.7	16.1	12.6	6.1	1.4	0.1			
Mean Daily Temperature Range	MDBR	9.6	8.8	10.1	8.8	7.9	7.7	8.5	8.7	8.2	5.9	7.1	8.8			
	5% DB	8.8	6.5	6.6	7.3	14.0	12.2	14.2	12.8	11.2	6.1	5.6	7.0			
	MCDBR	8.2	5.8	5.7	5.1	8.0	6.7	7.3	7.8	7.7	4.9	5.2	6.6			
	5% WB	8.8	6.8	6.4	6.7	13.4	11.3	12.9	9.3	5.7	5.7	6.9	6.9			
	MCWBR	8.3	6.2	5.7	4.8	7.8	6.6	7.3	7.2	6.7	4.7	5.4	6.5			
	MCWBR	8.3	6.2	5.7	4.8	7.8	6.6	7.3	7.2	6.7	4.7	5.4	6.5			
Clear-Sky Solar Irradiance	taub	N/A	0.252	0.302	0.329	0.351	0.347	0.361	0.343	0.311	0.263	N/A	N/A			
	taud	N/A	2.018	2.088	2.146	2.298	2.444	2.436	2.466	2.509	2.181	N/A	N/A			
	Ebn at Noon	0	441	678	791	825	846	817	788	714	436	0	0			
	Edh at Noon	0	41	81	105	104	93	91	78	56	34	0	0			
All-Sky Solar Radiation	RadAvg	0.00	0.30	1.53	3.34	4.52	4.78	4.35	2.97	1.54	0.44	0.00	0.00			
	RadStd	0.00	0.03	0.11	0.23	0.45	0.39	0.45	0.28	0.14	0.05	0.00	0.00			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page